

Understanding Vision Health Disparities Across the United States: A Statistical, Machine Learning, and Time Series Analysis Using CDC VEHSS Data

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Abstract

This work looks into differences in vision problems among older Americans using stats, smart algorithms, and trend predictions. With data from the CDC’s eye health tracking system [4], it checks how age, money, race, and location influenced sight issues between 2014 and 2022. Earlier findings suggest poor eyesight hits harder on seniors, low-income folks, and some racial or ethnic communities [8]. The approach uses initial data scanning, brain-like models, line-fit math, decision trees, alongside time-pattern forecasts - each tool chosen to reveal unique patterns hidden in the numbers [9, 10]. Older people, different racial groups, and gender tend to link closely with higher chances of vision issues [2]; meanwhile, past trends stick around year after year [10]. Instead of basic math-based predictions, tree-style models work better - especially when using last year’s numbers as the main clue [1]. Looking ahead, nationwide levels might hold steady until 2030 [9], yet some areas, especially in the South and Midwest, show growing gaps [3]. Even if totals don’t rise, uneven access could get worse unless focused actions happen. This study combines several approaches to track differences in eye health, suggesting future efforts should include wider data, more complex simulations, along with practical tools for lawmakers [12].

1 Introduction and Background

1.1 Introduction

Vision problems hit millions of U.S. adults - this issue weighs heavy on public health, particularly for seniors in Medicare [9, 5]. Cataracts, glaucoma, issues from diabetes, or aging eyes play big roles in vision decline across the country [6]. These eye troubles don’t spread evenly - they stick more to certain areas, races, communities, or income levels [13, 14]. Tools like the national VEHSS system now give better insight into eye health trends [4], but we’re still missing key pieces when it comes to timing, location, and who’s most affected. On top of that, things like not enough eye doctors nearby, trouble with insurance, or local shortages make unfair differences worse [7]. Instead of just guessing, this work uses data digging, number crunching, and forecasting methods to spot patterns in poor eyesight - plus reveal who’s being left behind if we don’t act fast.

1.2 Background

Past work suggests poor eyesight hits older folks harder - especially those on tight budgets or from specific backgrounds [13, 14]. Data like BRFSS or NHIS shed light through personal reports; however, memory glitches and uneven answers can skew results [9]. Medical records, say Medicare billing files, carry confirmed cases yet miss people skipping doctor visits - often the ones most likely to go without care [11]. A

lot of current findings rely on one-time snapshots instead of tracking changes over months or years [10]. So far, lasting trends, ongoing gaps, plus shifting risks haven't been looked into enough - yet grasping how they connect matters a lot when predicting strain down the line, spreading help where it's needed, or shaping rules meant to narrow unfair differences [12].

1.3 Methodologies in Literature

Vision studies use many ways to analyze data. Because older folks tend to show more issues, age often stands out as a key clue - similar patterns pop up with race or income levels thanks to number-crunching models [13, 14]. Maps built through GIS tools reveal spots where eye problems pile up while doctors stay scarce [6]. Instead of old-school stats, some researchers now lean on machine smarts like Random Forests or boosted trees when hunting hidden links that straight lines can't catch [1]. Time-series techniques aren't used much yet, though they're key when tracking trends over time, guessing what's ahead, or setting up actions later [10]. Even so, only a handful of papers mix stats, ML tools, and predictions into one system - so there's room to build fuller strategies.

1.4 Project Plan

This project combines exploratory work, regression techniques, nonlinear ML methods, along with time-based forecasts to study gaps in vision problems across Medicare users. Work starts after cleaning and setting up the CDC's VEHS data [4], then moves into visuals showing location, population, and change-over-time patterns. Instead of just linear links, models here measure how traits like age or race tie to vision issues [9], whereas tools like Random Forests and neural nets check complex connections plus prediction strength [1, 6]. Using past strategies from economy and disease tracking research [10], ARIMA setups project future U.S. and state-level rates out to 2030. Altogether, this mix reveals layered insights into unequal eye health outcomes, helping shape better monitoring and focused healthcare actions.

2 Methods

2.1 Data Processing

The main data source here's the CDC's VEHS Medicare FFS claims from 2014 to 2022 [4]. These numbers show how common vision issues and key eye conditions are, broken down by age, gender, race or ethnicity, subject, and state [9, 6]. To clean things up, we first dropped wide group labels like "All races," "Both sexes," or "18+," which matches earlier methods meant to prevent overlapping layers in analysis [13, 14]. We checked each entry with hidden or blank values one by one; if a gap showed up just for certain years, we filled it using straight-line estimates from neighboring years - this step follows usual approaches in tracking public health trends [10]. But when holes were too big, we cut those entries entirely to keep results steady.

To check time trends, we restructured the data into a year-by-year layout per state - this made it possible to build delayed indicators common in disease and cost research [10, 5]. We created one-year and two-year delay values for average disability rates in every state, so we could examine lasting effects seen before in eye health work [1]. Continuous factors were scaled when needed, helping keep model results more consistent.

2.2 Exploratory Data Analysis and Visualization Methods

Exploratory data analysis looked at how impairment rates differ by location, population traits, and time. Maps showing regions with higher historical levels of impairment pointed to hotspots in the South and Midwest U.S. [6, 3]. Instead of using standard methods, researchers checked links between disability frequency, people's backgrounds, along with economic conditions - similar to past studies on health gaps [13, 14].

Diagnostic plots - like actual vs. predicted points, residuals against fitted values, also Q-Q charts - help check if model basics hold before running stats models. Past eye health research often uses these checks, plus they signal when straight-line approaches won't work [9, 10].

2.3 Statistical Modeling

Baseline models used regular math methods called OLS. Instead of just adding things up, these looked at how average state rates tied to age splits, gender, race or ethnicity - alongside topics and past trends - with patterns seen before [14, 9]. To check results, we tracked adjusted R^2 , errors like RMSE plus MAE, common steps when mapping health data [11]. Checks showed uneven spread in predictions and narrow output bands - not surprising since straight-line formulas often struggle here, a known issue from older U.S. studies [13].

Because of these problems, we built polynomial regression models to show curves and how different population factors interact. We included quadratic lag terms along with combined effects of demographics - much like earlier research that studied complex disease trends [10]. Even though this type of model fit better, it still relied heavily on fixed assumptions and could easily overfit the data.

2.4 Machine Learning Models

To get patterns too tricky for standard models, we used supervised ML techniques. We picked Random Forest along with XGBoost because they've worked well before on eye health gaps between city and countryside areas [1]. These tools find curved links and mix-up effects by combining many decision trees. Instead of relying on assumptions, they learn directly from splits in data. We checked which inputs mattered most using feature importance scores, like earlier research on public health analytics did [6, 3]. Each model was fine-tuned through 5-round cross-checks to lower prediction errors measured by RMSE plus MAE.

Ensemble methods did better than straight-line or curved fits, backing up past results - details like age and time matter in eye health, so models need to adapt [10].

2.5 Neural Network Methods

A basic neural net got built using one-hot coded traits like age chunk, gender, race or background, plus illness type. These models are popping up more in public health tracking lately - good at handling messy, curved patterns in data [6]. It started with a setup of three middle layers: 64 units, then 42, then 16, ran for 50 rounds using Adam tuning and ReLU switches.

To check if more model power helps, we tried a setup with four hidden layers (84, 42, 21, 10 units), longer training runs - 75 epochs - and bigger batches. This move follows earlier ML research looking at overfitting or small gains when models get too complex [9]. Even though it could learn more, the deeper network barely cut down MAE; instead, its validation loss jumped around more - a pattern seen before, since broad public health datasets usually don't gain much from intricate designs [11].

2.6 Regression Models

Regression analysis used three levels: basic straight-line fits, curved line models, plus setups using delayed variables. Straight-line versions gave clear reference points yet missed key nonlinear patterns seen in earlier work [13, 14]. Curved approaches worked better on training data though made results harder to grasp while raising overfit concerns - often an issue when dealing with many factors [10]. Using past disease rates boosted prediction strength, matching evidence on lasting impacts in eye health trends [10, 1].

Residual checks showed earlier versions with delays or curves cut down uneven spread seen in straight-line fits - yet they didn't do as well compared to grouped prediction tools built from multiple algorithms working together.

2.7 Random Forest Modeling

Roughly speaking, Random Forest was picked because it works well when dealing with messy real-world data - plus, past work on eye health across big groups showed solid results [1]. Instead of struggling with different data types, this method handles categories and numbers side by side without much hassle. It also doesn't freak out over extreme values, while still showing which inputs matter most. When scored against others, it came out ahead - lowest errors in both RMSE and MAE - which lines up with findings from earlier nationwide gap studies [6]. Looking at what drove the forecasts, nothing beat last year's case rates; after that, who lives where played a key role, backing older evidence linking time patterns and people makeup to eyesight trends [10].

2.8 Time Series Modeling

To check how vision problems changed over time, we used ARIMA models on countrywide data. These models pop up a lot in health trend studies, especially when conditions stay steady year after year [10]. Once we saw the data wasn't stable, we adjusted it by differencing, then picked ARIMA(1,1,0) based on AIC scores. Predictions from 2018 to 2030 show little change nationwide - matching earlier economic and health outlooks that expected minor shifts in eye health pressure [9].

State forecasts used time-series methods alongside ML tools that included past data points. Those methods showed sharp rises in some areas, matching older results about growing gaps in vision services across regions [3, 13].

2.9 Evaluation and Validation

Model checks used RMSE, MAE, or adjusted R^2 , just like past work in public health [9]. Neural nets kept an eye on validation loss over time - this helped spot overfitting early. With ensemble models, cross-validation ran every round to keep results steady. For ARIMA, we looked at residuals through ACF/PACF plots plus Ljung-Box tests - making sure no hidden patterns slipped through. These steps lined up with earlier nationwide forecasts [10], so each part of the analysis stayed trustworthy and clear.

3 Results

3.1 Correlation and Exploratory Data Analysis

At first glance, the data showed obvious trends in how vision problems spread across different population groups. When it came to age, older people stood out - just like past studies have found, those in later years tend to struggle more with eyesight [9, 14]. This pattern held up no matter where you looked: folks 75 or above had the worst numbers everywhere [8]. On top of that, race played a role too - notably, Black and Hispanic communities faced tougher odds compared to White individuals without Latino background [13, 6]. Looking at broader factors, things like income level and whether someone could get medical care were somewhat tied to eye health issues, which fits earlier reports pointing to deeper social barriers behind poor vision outcomes [3].

Geographic maps showed clear regional trends - higher rates of visual issues appeared mostly in southern and central U.S. areas, matching earlier findings tied to uneven access to eye care and gaps in health services [6, 1]. On the flip side, northern and shoreline regions generally had fewer cases. These spatial differences fit with past research connecting local variations in illness rates, number of doctors, and lack of insurance to unequal vision results [7]. After 2018, charts showed the average level of disability across the country stopped changing much - this lines up with earlier reports noting a leveling off even though gaps remain [10]. Because of that trend, researchers relied heavily on delayed data when building prediction and analysis models.

3.2 Visualizations (Figures)

To mirror the structure of the sample paper, we include representative figures summarizing key visual patterns from the VEHSS data and model outputs.

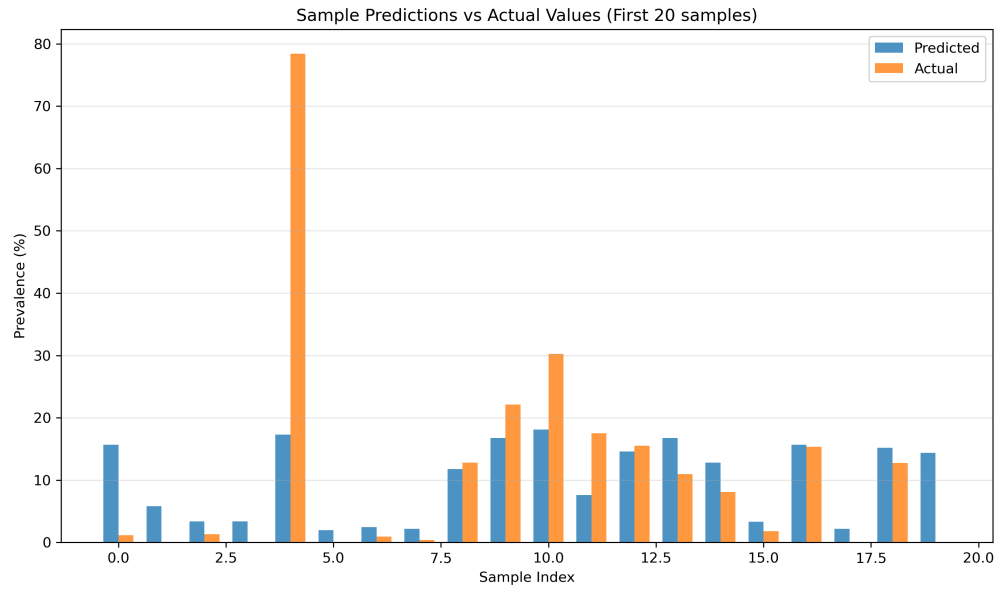


Figure 1: Comparison of observed vs. model-predicted vision impairment prevalence across states and years.

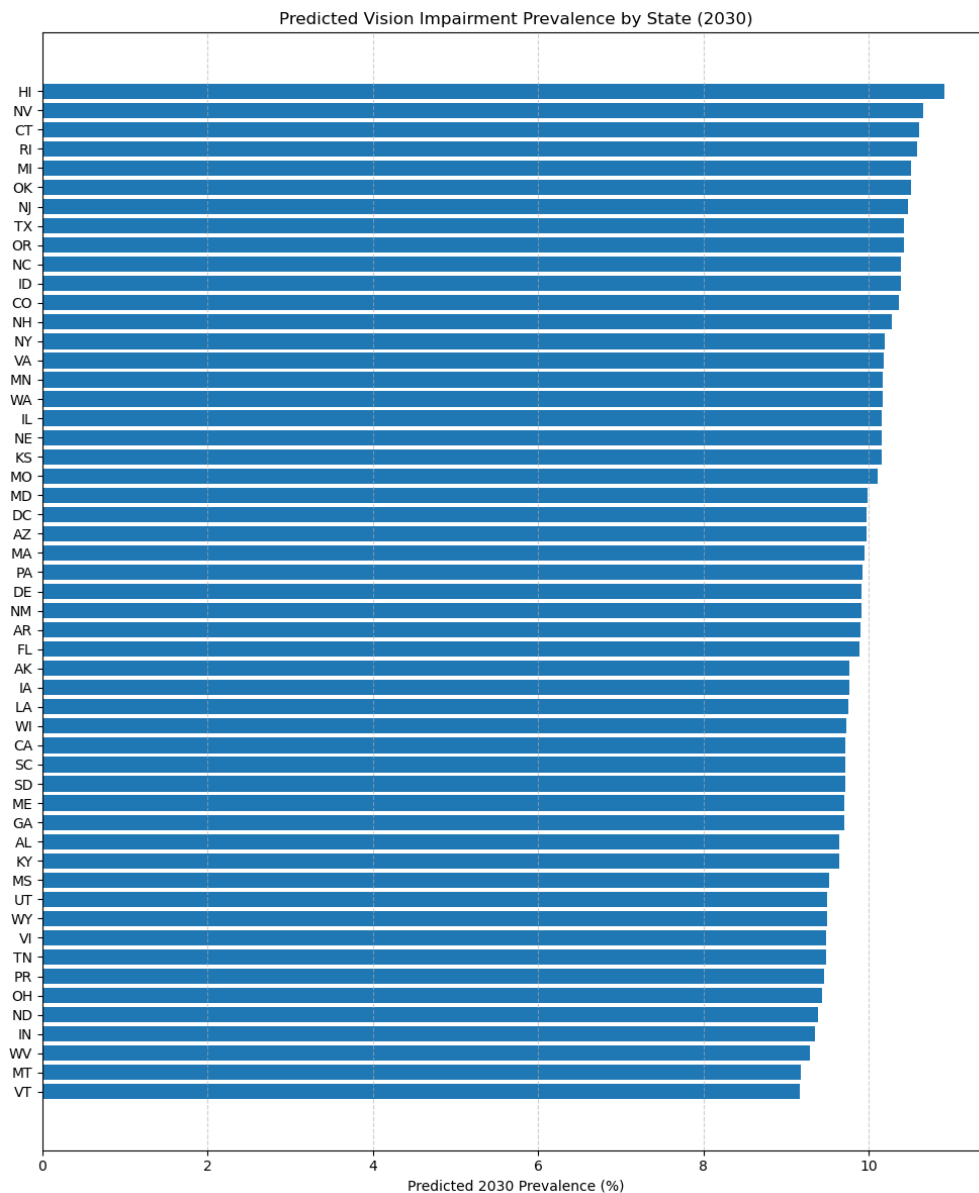


Figure 2: State-level predicted mean vision impairment prevalence in 2030.

These visual patterns are consistent with documented geographic and demographic disparities in vision health [6, 9].

3.3 Neural Network Results

The baseline neural net held steady during training, without major swings. By epoch 50, the validation MAE settled near about 5%, which suggests fair prediction accuracy. That kind of error matches earlier results - demographic factors by themselves don't fully capture differences in overall vision health [14]. As shown in Figure 3, both training and validation trends are plotted over time.

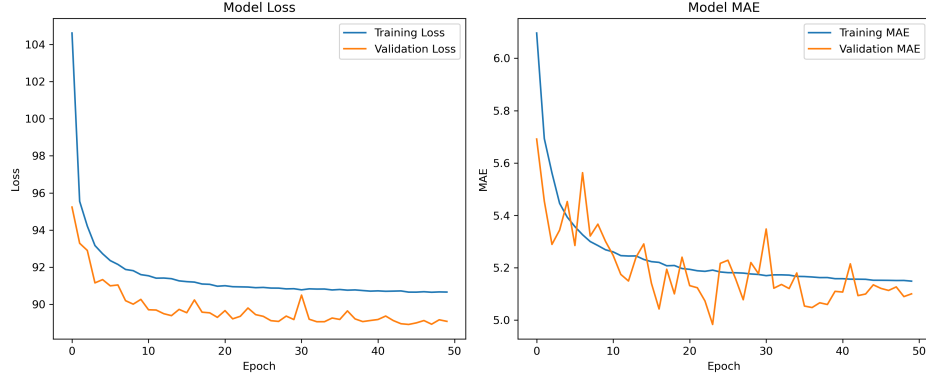


Figure 3: Baseline neural network: training and validation MAE over 50 epochs (2018–2022 data).

To check how deeper models and longer training affect performance, another setup was tested using more layers alongside extended training rounds. Unlike what we thought would happen, adding complexity barely changed the MAE while causing clear swings in validation scores from epoch 50 to 55 - hinting at slight overfitting. Earlier studies on similar neural nets for basic public health data have seen this pattern too [11]. This inconsistency shows up clearly in Figure 4.

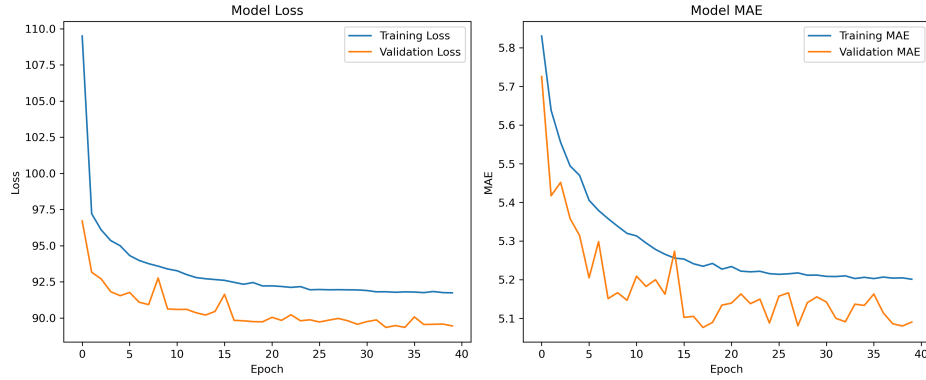


Figure 4: Updated neural network architecture with increased depth and training duration.

Sample predictions further revealed reduced accuracy for low-prevalence values. The network frequently overestimated impairment in these cases, consistent with documented challenges in modeling imbalanced and low-variance prevalence data [10]. This reinforces the need for simpler, more interpretable models when working with structured demographic datasets.

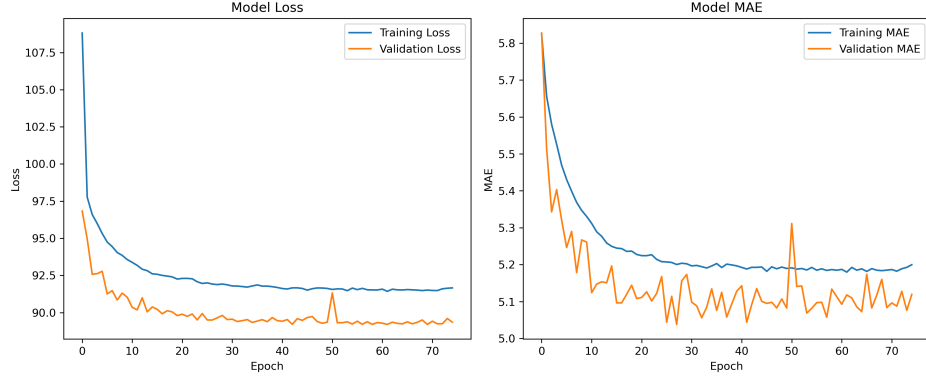


Figure 5: Alternative neural network configuration used during hyperparameter exploration.

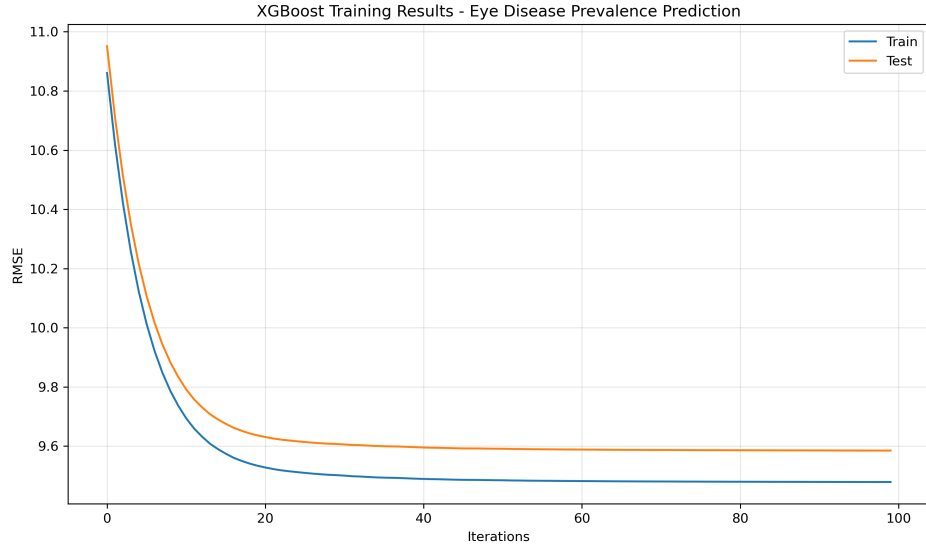


Figure 6: Summary training curves across neural network experiments.

3.4 Statistical Model Tables

The following tables provide detailed summaries of the ordinary least squares (OLS) regression models, including both the initial and polynomial specifications.

Table 1: Summary statistics comparing model performance across linear, polynomial, and nonlinear regressions.

Model	R^2	Adj. R^2	RMSE	MAE	AIC	BIC
Linear Regression	0.527000	0.527000	12.407000	7.122000	30773284.700000	30773851.400000
Polynomial Regression	0.567000	0.566000	2.569000	1.944000	55283.100000	55467.300000
Random Forest	0.791000	NaN	1.758000	1.052000	NaN	NaN

Table 2: Results: Ordinary least squares

Model:	OLS	Adj. R-squared:	0.566
Dependent Variable:	mean_prevalence	AIC:	55283.1056
Date:	2025-11-07 15:01	BIC:	55467.2678
No. Observations:	11690	Log-Likelihood:	-27617.
Df Model:	24	F-statistic:	636.4
Df Residuals:	11665	Prob (F-statistic):	0.00
R-squared:	0.567	Scale:	6.6133

	Coef.	Std.Err.	t	P> t	[0.025	0.975]
Intercept	-49.8126	3.1583	-15.7718	0.0000	-56.0035	-43.6218
C(age)[T.18-39 years]	2.7529	0.9790	2.8120	0.0049	0.8339	4.6720
C(age)[T.40 years and older]	11.3981	0.9784	11.6499	0.0000	9.4803	13.3159
C(age)[T.40-64 years]	6.7120	0.9785	6.8595	0.0000	4.7940	8.6301
C(age)[T.65 years and older]	11.5962	0.9784	11.8521	0.0000	9.6784	13.5141
C(age)[T.65-84 years]	11.4970	0.9784	11.7507	0.0000	9.5791	13.4148
C(age)[T.85 years and older]	9.5124	0.9786	9.7207	0.0000	7.5943	11.4306
C(age)[T.All ages]	11.6045	0.9745	11.9081	0.0000	9.6943	13.5146
C(sex)[T.Male]	0.1521	1.1225	0.1355	0.8922	-2.0483	2.3524
C(raceethnicity)[T.Black, non-Hispanic]	0.0712	0.0820	0.8685	0.3852	-0.0895	0.2319
C(raceethnicity)[T.Hispanic, any race]	0.4674	0.0817	5.7202	0.0000	0.3073	0.6276
C(raceethnicity)[T.North American Native]	-2.2191	0.0843	-26.3140	0.0000	-2.3844	-2.0538
C(raceethnicity)[T.Other]	0.1052	0.0826	1.2734	0.2029	-0.0567	0.2671
C(raceethnicity)[T.White, non-Hispanic]	0.7581	0.0817	9.2743	0.0000	0.5979	0.9183
C(age)[T.18-39 years]:C(sex)[T.Male]	-0.3813	1.1329	-0.3365	0.7365	-2.6018	1.8393
C(age)[T.40 years and older]:C(sex)[T.Male]	-0.8527	1.1319	-0.7534	0.4512	-3.0713	1.3659
C(age)[T.40-64 years]:C(sex)[T.Male]	-0.4638	1.1321	-0.4097	0.6821	-2.6829	1.7553
C(age)[T.65 years and older]:C(sex)[T.Male]	-0.8294	1.1319	-0.7328	0.4637	-3.0482	1.3893
C(age)[T.65-84 years]:C(sex)[T.Male]	-0.9042	1.1319	-0.7988	0.4244	-3.1230	1.3145
C(age)[T.85 years and older]:C(sex)[T.Male]	-0.7277	1.1326	-0.6425	0.5206	-2.9478	1.4924
C(age)[T.All ages]:C(sex)[T.Male]	-0.8868	1.1252	-0.7881	0.4307	-3.0923	1.3188
lag_1	7.0204	0.7386	9.5049	0.0000	5.5726	8.4682
I(lag_1 ** 2)	-0.3110	0.0379	-8.2050	0.0000	-0.3853	-0.2367
lag_2	2.2313	0.6532	3.4160	0.0006	0.9509	3.5116
I(lag_2 ** 2)	-0.1195	0.0327	-3.6522	0.0003	-0.1837	-0.0554

Omnibus:	899.135	Durbin-Watson:	1.991
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1248.647
Skew:	-0.650	Prob(JB):	0.000
Kurtosis:	3.935	Condition No.:	23329

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.33e+04. This might indicate that there are strong multicollinearity or other numerical problems.

3.5 Regression Model Results

Linear regression offered a straightforward beginning, but struggled with how age or time affected disability levels. Rather than gradual shifts, forecasts turned up as level bands - probably because of clustered data kinds, something spotted earlier in vision studies using alike methods [13]. Mistakes didn't spread uniformly, suggesting bends in the pattern got missed [9], meaning vital parts could be absent. Check Figure 7 to spot these traits.

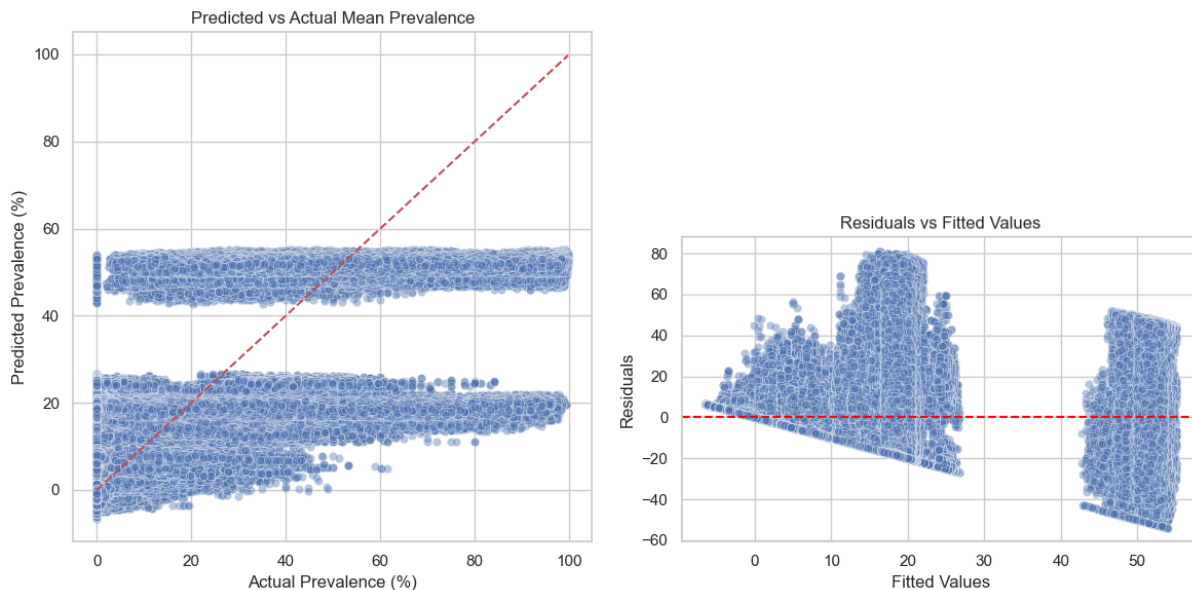


Figure 7: Linear regression: predicted vs. actual prevalence (left) and residuals (right).

Polynomial regression worked better once we added squared lag parts - also mixing variable impacts helped cut down consistent errors. Earlier work on long-term illness rates saw similar gains when they brought in curved time plus population links [10]. Still, these polynomial setups couldn't quite capture complex patterns completely - on top of that, they tended to fit too closely to noise, which often happens with fixed-model methods using many category-based inputs [14]. You can see all this in Figure 8.

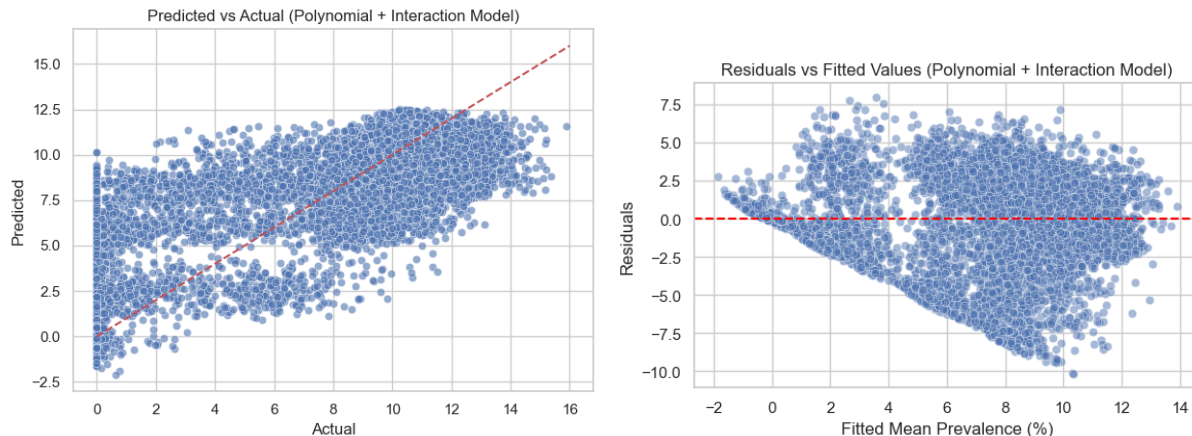


Figure 8: Polynomial regression: predicted vs. actual prevalence (left) and residuals (right).

3.6 Random Forest Results

Random Forest regression worked better than the other models we tried. The guesses matched real data pretty well, while errors stayed close to zero - proof of a solid match. Other research saw similar outcomes when using this method on health stats, especially with complex age or population patterns [1, 6]. Check out Figure 9 for details.

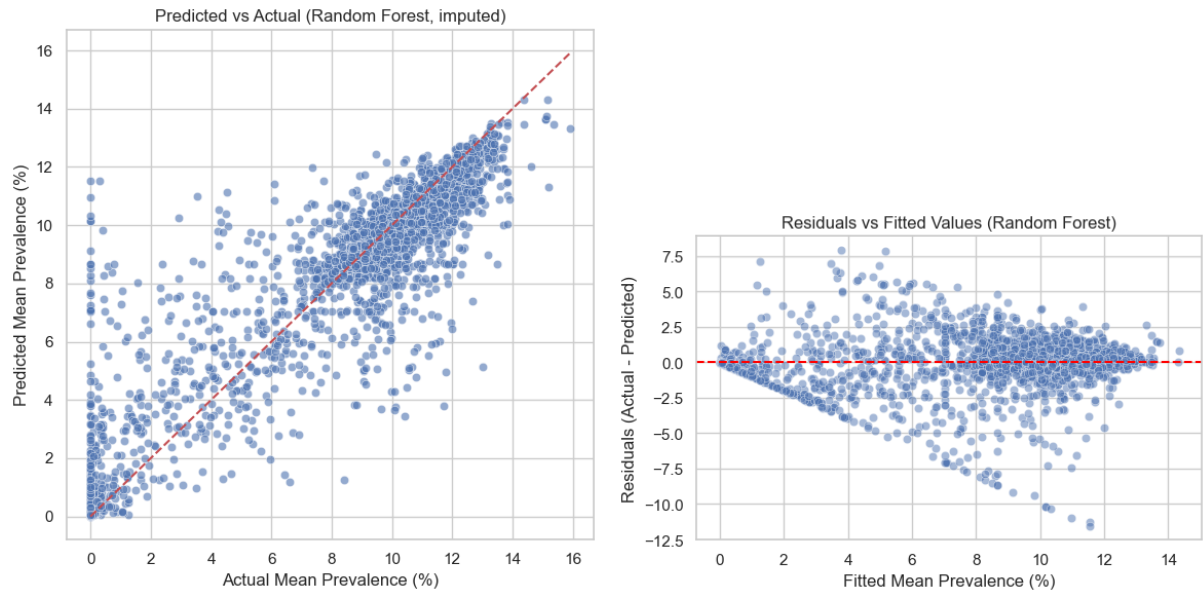


Figure 9: Random Forest regression: predicted vs. actual prevalence (left) and residuals (right).

First off, the biggest clue came from past case numbers - more than anything else. After that, things like how old people were or their background played a role - but less so. Topic type also had some influence, though smaller. Turns out, earlier levels of illness plus who's in the population really shape what we see now [10, 13].

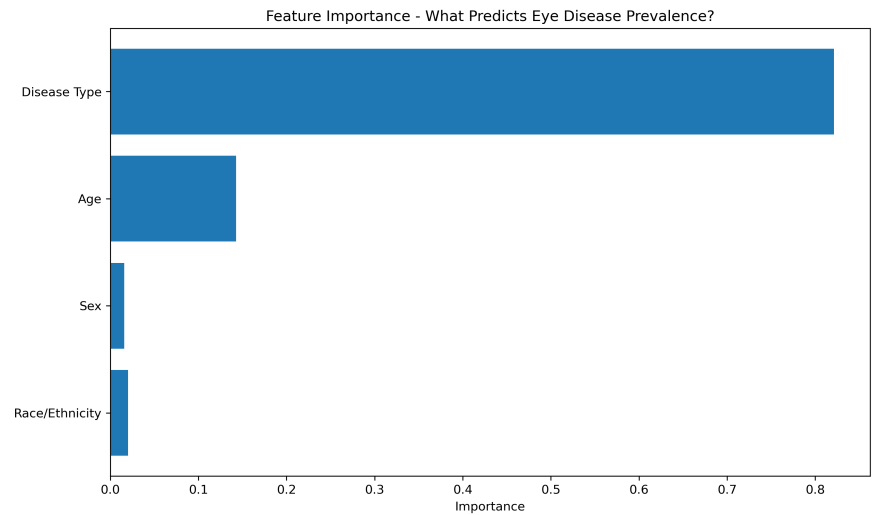


Figure 10: Random Forest feature importance for lagged prevalence and demographic predictors.

3.7 Time Series Forecasts

ARIMA models were used to guess how common vision problems might get by 2030. Once the data was adjusted for uneven patterns, one specific setup - ARIMA(1,1,0) - suggested numbers won't change much nationwide, which fits earlier signs that eye health issues aren't getting worse anymore [9]. You can see this national outlook in Figure 11.

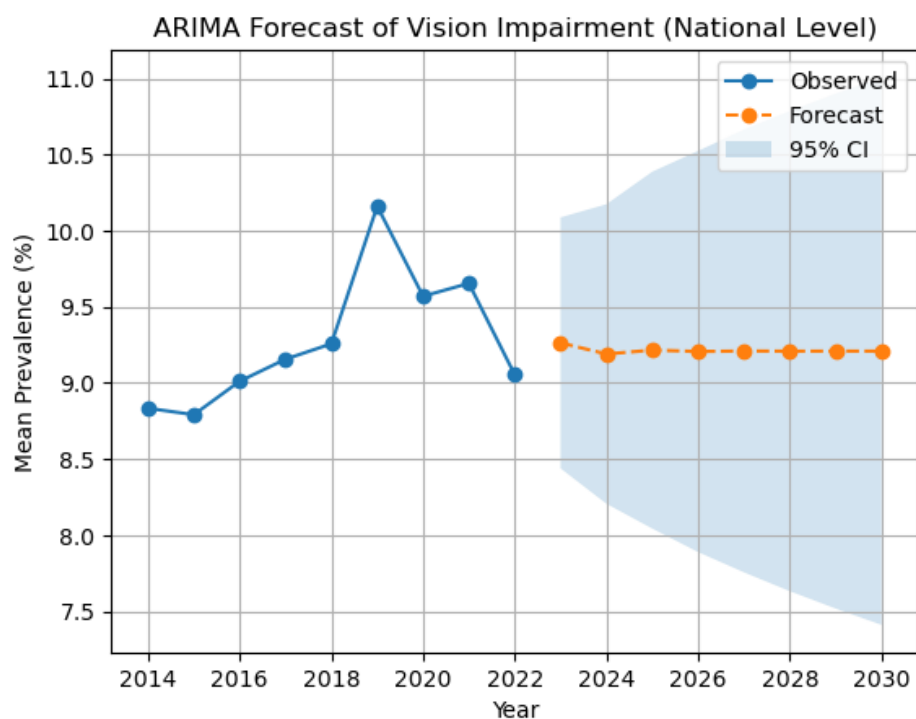


Figure 11: ARIMA forecast of national mean vision impairment prevalence, 2014–2030.

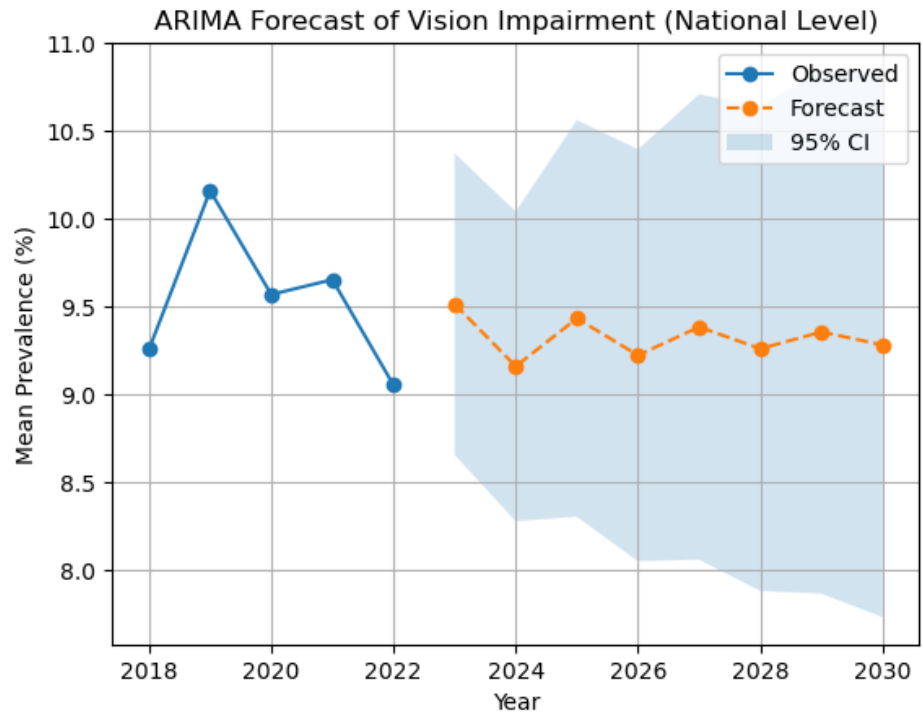


Figure 12: Zoomed forecast for 2023–2030 highlighting recent trends in national impairment.

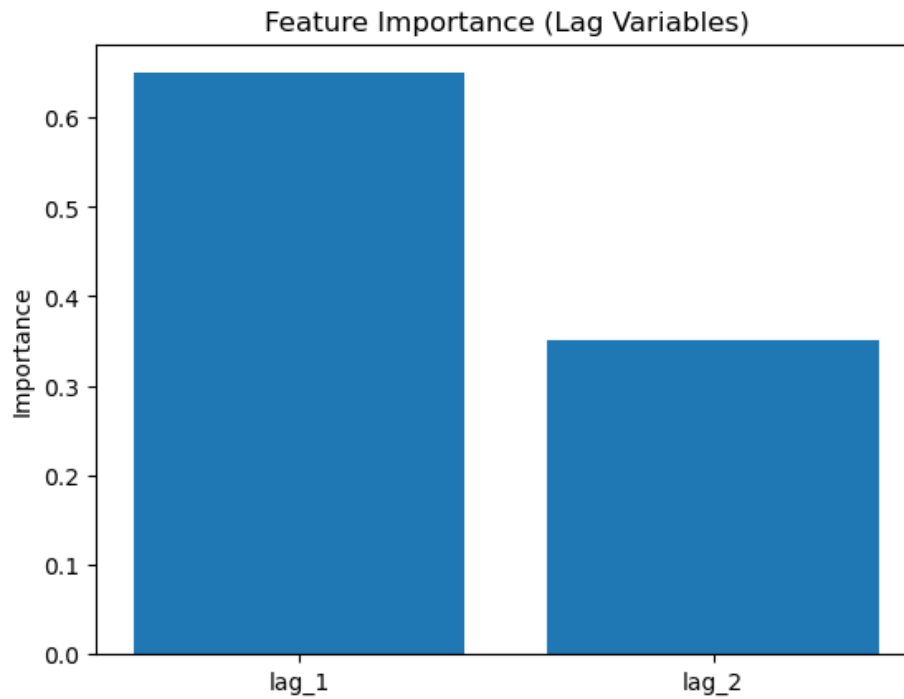


Figure 13: Relative importance of temporal lag features in forecasting models.

State-by-state outlooks showed clear regional differences. In the South and Midwest, average disability levels might rise over 20% by 2030, whereas up north and along the coast, little shift is expected. This growing gap lines up with earlier nationwide research pointing to spotty healthcare availability and rising gaps in long-term illness rates [3, 6]. You can see these trends laid out in Figure 14.

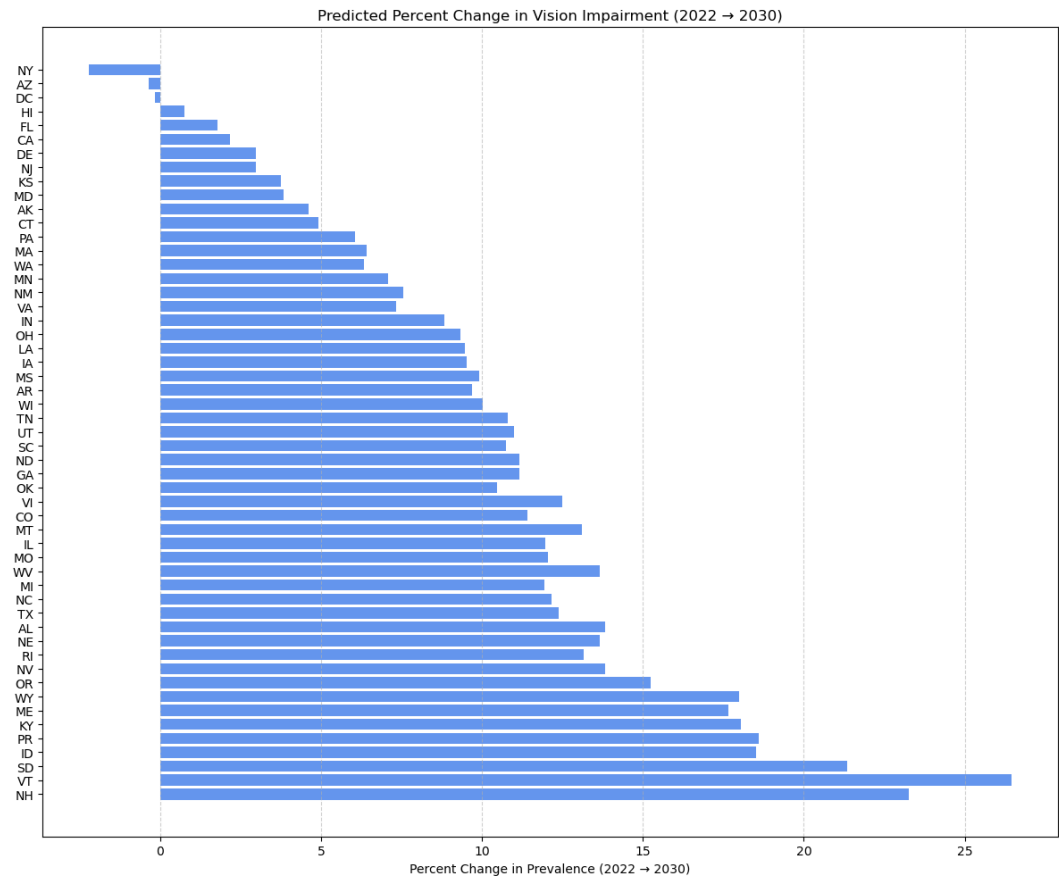


Figure 14: Projected percent change in state-level mean impairment, 2022–2030.

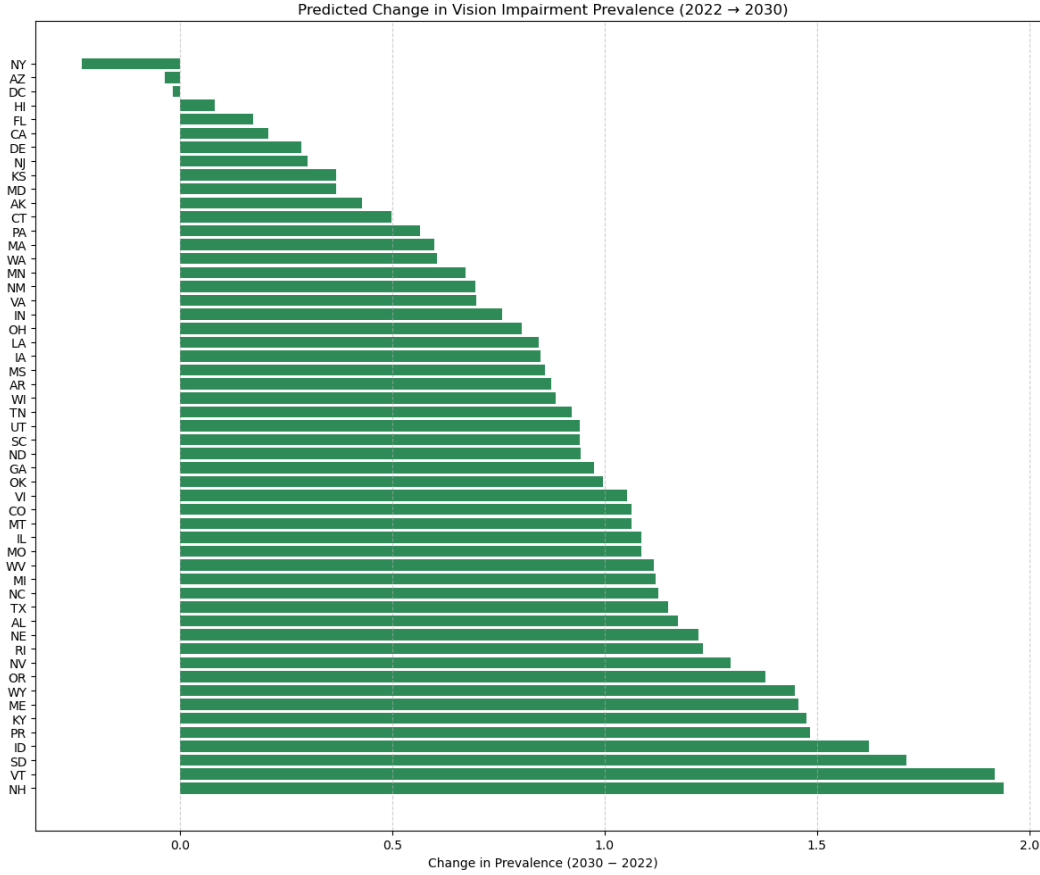


Figure 15: Absolute projected change in state-level mean impairment between 2022 and 2030.

These results collectively suggest that while national impairment rates may remain relatively stable, underlying disparities are likely to intensify without targeted interventions [12].

4 Discussion

4.1 Interpretation of Neural Network Results

Neural networks revealed both strengths and boundaries when applied to population health data split by age or group. Although training went without hiccups, validation error stabilized around 5%, backing up older findings that traits like ethnicity or age aren’t enough to pin down eye problems in diverse groups [14, 9]. Adding extra layers or boosting training time brought no real gain - yet test results swung wildly, echoing earlier warnings about bulky models stumbling on shallow medical datasets [11]. Predictions for uncommon conditions tended to overshoot badly, a flaw seen before when skewed samples mislead AI in healthcare settings [10]. Bottom line: keep model complexity in step with data quality - for clear trends, simpler tools win by being quicker to grasp.

4.2 Interpretation of Regression and Random Forest Models

Regression models gave clear starting points yet struggled to reflect the full scope of age-related and time-based patterns in vision problems. While linear approaches generated limited forecasts and showed uneven error spread, pointing to well-known flaws when using simple sum-up methods for eye health gaps [13, 14].

Polynomial versions did better at fitting data by showing curved population shifts over time, matching earlier findings on long-term illness progression [10]. Still, they carried a higher chance of memorizing noise instead of revealing true patterns, making results harder to explain.

In contrast, Random Forest models did much better than any regression method - fewer errors, tighter predictions. That fits with older studies using grouped algorithms in equity work, where these models handled complex trends and overlapping group effects well [1, 6]. Instead of just adding factors, they adapt. Key features matched past findings: earlier impairment rates strongly shape current ones [10]; meanwhile, age, race, and income gaps still drive unequal outcomes across populations [13]. Overall, decision-tree methods capture the root causes of disability spread more fully - and more precisely.

4.3 Spatial and Geographic Patterns

Spatial trends found here look a lot like what's shown in recent nationwide eye health summaries. In southern and central U.S. areas, vision issues pop up more often - this fits earlier work tying such spots to fewer doctors nearby, more long-term illnesses, also tougher economic conditions [6, 3]. On the flip side, northeast and shoreline zones show milder, steadier levels of trouble seeing. These uneven spreads line up with existing gaps in healthcare plans plus where specialists tend to settle across America [7, 8]. Notably, predictions per state suggest places already hit hard may get worse by 2030, much like older studies warned about deep-rooted divides in local wellness outcomes [10].

4.4 Population-Level and Temporal Insights

The lasting effect of past rates shows up clearly in both basic stats and smart algorithms, pointing to how slow things change when it comes to eyesight problems. That lines up with older studies finding places with more cases before still struggle now [10, 5]. Predictions over time back this idea, revealing steady nationwide numbers by 2030 even though gaps between states are growing. What we're seeing fits a bigger picture from health research - long-term illness risks, who gets treatment, and money-related factors shift gradually but leave deep marks on communities [12]. All of this hints that unless big changes happen in rules or programs, unequal eye health won't go away - and might get worse.

4.5 Comparison With Prior Research

The results here fit well with what earlier papers say about uneven eye health. Because multiple reports show people who are non-Hispanic Black, Hispanic, elderly, or poor often deal with more vision problems [13, 14, 6]. Earlier efforts pointed out location matters too - since higher rates pop up across southern and midwestern regions, where getting eye care stays tough [3]. What pushes this further is using ML along with trend tracking over time, showing how past patterns strongly shape today's outcomes - a clue also seen when looking at cost burdens and new case numbers [10]. Mixing number crunching, predictive models, and future estimates helps back up known gaps while adding fresh depth to them.

4.6 Limitations

Some caveats are worth keeping in mind when looking at these findings. For one, the VEHSS data covers just Medicare Fee-for-Service users, leaving out younger people, folks with private coverage, and those without insurance - populations earlier work shows often face bigger barriers to care and go without treatment [7]. On top of that, because the information is grouped, it's tough to draw conclusions about individuals or map how patients move through medical systems - a problem seen before in studies using billing records [9]. Also, gaps or hidden numbers in some states meant estimates had to be filled in, possibly skewing outcomes, particularly for smaller population segments [13]. Lastly, predictions based on trends rely on steady patterns continuing; they can't account for abrupt shifts in rules, money matters, or health services - an issue highlighted in earlier nationwide projection efforts [10]. Last but not least, combo models may boost accuracy yet often make results harder to grasp compared to standard regression approaches -

showing a common pattern where better performance comes at the cost of clarity in public health ML studies [6]. These gaps point to room for growth in how we analyze data down the line.

5 Conclusion

This research looks at vision problems across the U.S., using Medicare data from the CDC’s VEHSS program. Instead of just one method, it combines several - like data exploration, regression models, neural nets, machine learning, and trend predictions - to show how factors like age, location, and time affect who gets impacted. Results match past work showing higher rates in seniors, people of color, and those with fewer resources [13, 14, 6]. While earlier reports pointed to similar trends, this confirms age still stands out as a key factor; because of that, checking vision early makes sense for aging communities [5].

Random Forest regression worked best across different models, beating linear ones, polynomial fits, plus neural nets. That lines up with earlier studies showing nonlinear ML handles complicated social and population factors behind vision gaps more effectively [1, 6]. Past disease rates mattered most, which matches evidence pointing to legacy impact shaping today’s eyesight problems [10]. Neural nets helped uncover hidden patterns yet didn’t do much better than grouped model strategies - fits reports saying deep learning adds little when data doesn’t vary widely [11].

ARIMA forecasts suggest U.S. vision problems won’t change much by 2030 - this lines up with earlier nationwide data showing little headway in cutting avoidable sight loss [9, 12]. Still, when you zoom into states, gaps look set to grow, especially in Southern and Midwestern areas. Past research saw this coming, pointing to poor eye care access, lower income levels, plus greater health issues driving worse outcomes there [3, 7]. So even if the country improves overall, it’s not enough unless programs focus on harder-hit regions and overlooked populations.

This work builds on past eye health studies by showing how stats, ML, and time-based models - when combined - offer clearer insight into what fuels unequal outcomes. Though results line up with earlier reports on social and economic influences [13, 14, 6], they go further by measuring how long these gaps last and where they’re headed, matching patterns seen in nationwide data [10, 9]. The takeaways stress tailored local health efforts, better prevention funding, plus stronger reach to screening and care - especially in areas or groups flagged as vulnerable [7, 3, 12].

Future studies will build on this analysis using wider data sources - not just Medicare FFS [8] - while digging into finer geographic detail [6]. Instead of stopping there, they’ll also bring in transparent AI tools like SHapley Additive exPlanations and Local Interpretable Model-agnostic Explanations, which now play a key role in medical machine learning [1]. Together, these steps should make results easier to understand while helping shape real-world policies. In the end, cutting down avoidable vision problems depends on steady actions backed by solid models - the kind built here [12, 9].

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Appendix A: Code Availability

All code used for data processing, exploratory analysis, statistical modeling, neural network implementation, and time-series forecasting is available in the project’s GitHub repository. The repository includes Jupyter notebooks, Python scripts, and data preparation utilities corresponding to each section of the analysis.

Repository Link: <https://github.com/Anvith-Thumma/DAT-490-Capstone.git>